

# Quantitative Trading Strategy Report

LGBM, GRPO and GARCH-Bollinger Strategies on HSI Futures

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# 1 Introduction

This project studies whether intraday Hang Seng Index (HSI) futures prices contain exploitable short-term patterns after transaction costs and execution frictions. The theoretical starting point is the Efficient Market Hypothesis (EMH). Under weak-form and semi-strong-form efficiency, prices should already reflect historical prices and public information, so persistent abnormal returns should be difficult to obtain. In practice, intraday markets can still display temporary predictability because of liquidity imbalance, volatility clustering, investor overreaction, market microstructure noise, and delayed information diffusion across day and night sessions.

The project compares three trading approaches. The first is a supervised learning strategy based on LightGBM (LGBM), which predicts next-bar returns from engineered price, volatility, volume, and calendar features. The second is a reinforcement learning strategy based on a GRPO-style policy optimization framework, where a policy network learns whether to be short, flat, or long from trading rewards. The third is a traditional GARCH-Bollinger strategy, which combines Bollinger Bands with GARCH conditional volatility, volume confirmation, and session-specific parameter selection.

The day and night sessions are separated because HSI futures do not trade under one homogeneous intraday environment. The day session overlaps with the cash equity market and usually has deeper liquidity and more local market participation. The night session is more exposed to overseas information, thinner liquidity, wider effective spreads, and larger gap risk. A signal that works during regular trading hours may therefore behave differently at night. The project keeps the full original dataset as the main market view, but also evaluates day-only and night-only datasets to test whether session-specific calibration improves robustness.

The report starts from market structure and trading rationale, then defines strategy rules, describes in-sample and out-of-sample design, presents parameter optimization, and finally discusses backtesting results, slippage, risk management, and implementation difficulties.

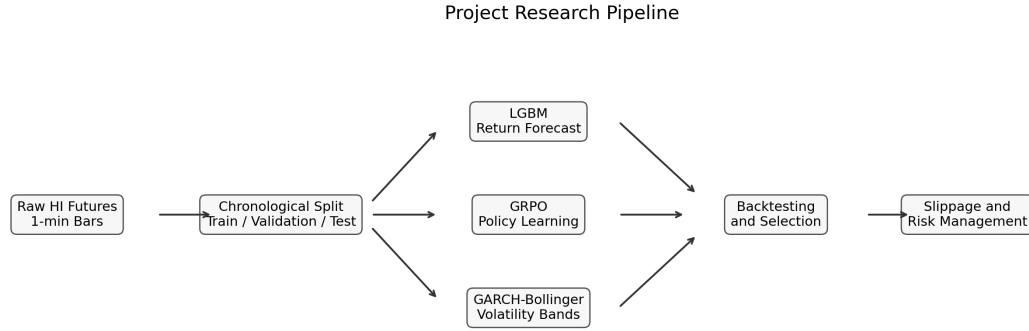


Figure 1: Research pipeline from processed HSI futures data to model training, backtesting, risk management and final comparison.

## 2 Trading Algorithms and Rationale

### 2.1 LGBM Supervised Learning Strategy

The LGBM strategy treats trading as a short-horizon return prediction problem. The prediction target is the next-minute close-to-close return:

$$y_t = \frac{P_{t+1}}{P_t} - 1. \quad (1)$$

The model is a LightGBM gradient boosting regressor. It is suitable for this task because tree boosting can model nonlinear interactions among technical indicators, volume variables, volatility measures, and time-of-day effects without requiring a linear return model.

The feature set contains 41 variables. They include lagged returns, log returns, rolling volatility, volume moving averages and volume ratios, candlestick shape variables, moving averages, moving-average deviations, moving-average crosses, RSI-style variables, and intraday calendar encodings. The feature design follows several economic intuitions:

- Lagged returns and moving-average deviations capture momentum and reversal.
- Rolling volatility captures volatility clustering.
- Volume ratios capture liquidity and market participation.
- Hour, minute-of-day and day-of-week encodings capture intraday seasonality.

The trading rule converts predicted returns into positions only when the expected return is large enough to cover round-trip cost:

$$q_t = \begin{cases} +1, & \hat{y}_t > c_t, \\ -1, & \hat{y}_t < -c_t, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where  $q_t$  is the target position and  $c_t$  is the transaction-cost and slippage threshold. This cost-aware threshold is important because a small positive return forecast may not be tradable after execution frictions.

## 2.2 GRPO Reinforcement Learning Strategy

The GRPO strategy formulates intraday trading as a sequential decision problem. The trading environment has three discrete actions:

$$a_t \in \{0, 1, 2\}, \quad 0 = \text{short}, \quad 1 = \text{flat}, \quad 2 = \text{long}. \quad (3)$$

The environment maps these actions to target positions  $\{-1, 0, +1\}$ . At each step, the environment calculates futures PnL using the current and next bar prices:

$$\text{GrossPnL}_t = q_t(P_{t+1} - P_t)M, \quad (4)$$

$$\text{Cost}_t = |q_t - q_{t-1}|(C + sM), \quad (5)$$

$$\text{NetPnL}_t = \text{GrossPnL}_t - \text{Cost}_t, \quad (6)$$

$$r_t = \frac{\text{NetPnL}_t}{\text{InitialCapital}}, \quad (7)$$

where  $M = 50$  is the HSI futures multiplier,  $C$  is fixed transaction cost per contract, and  $s$  is slippage in index points.

The observation vector contains standardized market features and the current position. Feature means and standard deviations are computed from the training set and saved with the model. The same normalization is then reused during validation, testing, and risk management, which avoids distribution leakage from the test set.

The optimization follows a GRPO-style relative reward idea. For a group of sampled episodes, the advantage of episode  $i$  is:

$$A_i = \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \quad (8)$$

where  $R_i$  is the total episode return. The policy loss is:

$$\mathcal{L} = -\frac{1}{N} \sum_i A_i \sum_t \log \pi_\theta(a_{i,t}|s_{i,t}) - \lambda H(\pi_\theta). \quad (9)$$

This design rewards policies that outperform other sampled trajectories in the same group. The entropy term encourages exploration and reduces premature convergence.

### 2.3 GARCH-Bollinger Strategy

The GARCH-Bollinger strategy is an interpretable benchmark. A standard Bollinger Band uses a moving average and rolling standard deviation:

$$\text{MA}_t = \text{RollingMean}(P_t, w)_{t-1}, \quad (10)$$

$$\text{Upper}_t = \text{MA}_t + k\sigma_t^{\text{roll}}, \quad (11)$$

$$\text{Lower}_t = \text{MA}_t - k\sigma_t^{\text{roll}}. \quad (12)$$

The one-period shift is essential: the band used for the decision at time  $t$  must be known before observing the trading result at time  $t$ .

The limitation of the standard Bollinger Band is that rolling standard deviation is a fixed-window historical estimate. During volatility clustering, it can react too slowly: after volatility suddenly increases, the window may still contain many low-volatility bars and the band can be too narrow; after volatility falls, old extreme bars may keep the band too wide. This directly affects trading because the strategy may generate too many signals in turbulent periods and too few signals after the market calms down.

The project extends the band width using GARCH(1,1) conditional volatility:

$$r_t = \sigma_t z_t, \quad (13)$$

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (14)$$

Since  $\sigma_t$  is return volatility, it is converted back to price scale:

$$\text{BandWidth}_t = kP_{t-1}\sigma_t. \quad (15)$$

Compared with rolling standard deviation, GARCH updates volatility through the recent return shock and the previous conditional variance. The  $\alpha$  term reacts to the latest shock, while the  $\beta$  term captures volatility persistence. This is the main rationale for the GARCH-Bollinger extension: the band width can adapt to volatility clustering more directly than an equal-weight rolling window.

Two signal interpretations are tested. The contrarian version treats a band breach as temporary overreaction and trades against the move. The momentum version treats it as a breakout and trades with the move. Some variants add volume confirmation:

$$\text{VolumeOK}_t = I(V_t > \rho \cdot \text{RollingMean}(V_t, w_v)_{t-1}). \quad (16)$$

### 3 Data Splitting and Experimental Design

The project uses processed HSI futures one-minute bars from July 2017 to June 2020. It constructs three datasets: the full original dataset, the day-session dataset, and the night-session dataset. Day session includes 09:15-12:00 and 13:00-16:30. Night session includes 17:15 to the next day 03:00.

All datasets are sorted chronologically and split into train, validation, and test sets with a 3:1:1 ratio. Chronological splitting is necessary because random splitting would allow future regimes to leak into the training process.

The project does not create separate bull-market and bear-market subsamples. The sample period contains an earlier upward phase and a later flatter or more volatile phase, but the group treats the full 2017-2020 period as one continuous HSI futures trading sample rather than manually classifying market regimes. This is a simplifying assumption. It avoids adding subjective regime labels and keeps the train-validation-test design consistent across LGBM, GRPO, and GARCH-Bollinger, but it also means that regime robustness is assessed indirectly through out-of-sample performance rather than through a separate bull/bear regime experiment.

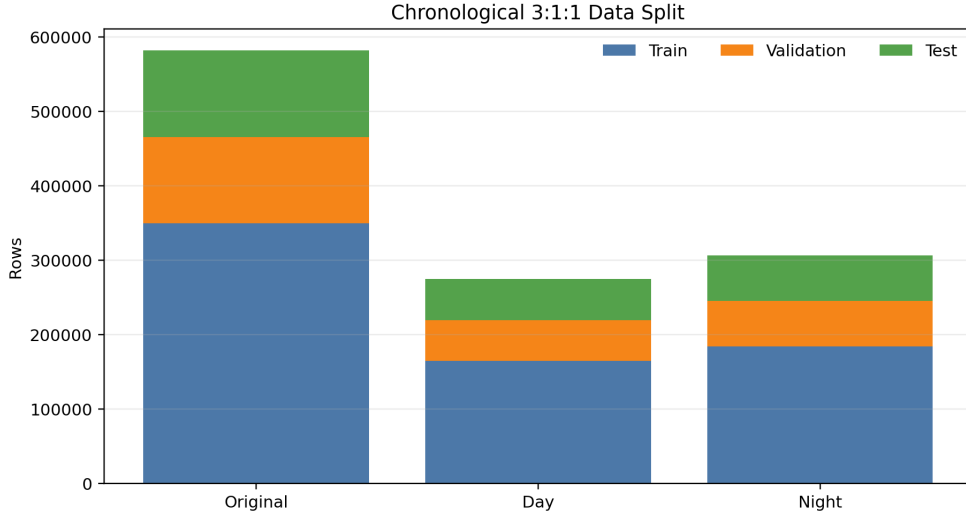


Figure 2: Chronological train-validation-test split for original, day and night datasets.

Table 1: Dataset split summary.

| Dataset  | Rows   | Train  | Validation | Test   | Start      | End        |
|----------|--------|--------|------------|--------|------------|------------|
| Original | 582100 | 349260 | 116420     | 116420 | 2017-07-03 | 2020-06-09 |
| Day      | 274448 | 164668 | 54890      | 54890  | 2017-07-03 | 2020-06-09 |
| Night    | 306133 | 183679 | 61227      | 61227  | 2017-07-03 | 2020-03-16 |

The roles of the three periods are:

- Train: fit model parameters and feature normalization.
- Validation: select models, scenarios, and trading parameters.
- Test: final out-of-sample evaluation only.

## 4 Parameter Optimization and Hyperparameter Selection

### 4.1 LGBM Hyperparameter Selection

The LGBM script uses five-fold `TimeSeriesSplit`. This preserves time ordering inside cross-validation and is more appropriate than shuffled K-fold validation for financial data. The objective is validation RMSE of next-minute return prediction.

The best parameters selected for all three datasets are:

Table 2: Selected LGBM hyperparameters.

| Hyperparameter                 | Value |
|--------------------------------|-------|
| <code>n_estimators</code>      | 500   |
| <code>max_depth</code>         | 5     |
| <code>num_leaves</code>        | 31    |
| <code>learning_rate</code>     | 0.02  |
| <code>min_child_samples</code> | 100   |
| <code>subsample</code>         | 0.90  |
| <code>colsample_bytree</code>  | 0.90  |

These settings are conservative for high-frequency return prediction. Shallow trees and larger minimum child samples reduce overfitting, while a low learning rate allows gradual boosting.

### 4.2 GRPO Training Parameters

The GRPO experiment uses 30 epochs, group size 8, 2048 steps per sampled episode, learning rate  $3 \times 10^{-4}$ , entropy coefficient 0.01, maximum gradient norm 1.0, hidden size 64, and random seed 42. These settings are used as a practical baseline rather than a fully optimized global search because reinforcement learning training is noisy and computationally expensive.

### 4.3 GARCH-Bollinger Parameter Optimization

The GARCH-Bollinger strategy uses validation-only grid search. The grid includes bar frequency, moving-average window, band multiplier, maximum holding bars for momentum strategies, volume window, and volume ratio. The tested values are:

Table 3: GARCH-Bollinger parameter grid.

| Parameter             | Values            | Role                                                                                                                        |
|-----------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Bar frequency         | 5, 15, 20 minutes | Resampled bar interval.                                                                                                     |
| Moving-average window | 20, 40, 60 bars   | Center line of the Bollinger band.                                                                                          |
| Band multiplier $k$   | 1.5, 2.0, 2.5     | Controls band width. Smaller $k$ creates narrower bands and more signals; larger $k$ creates wider bands and fewer signals. |
| Maximum holding bars  | 5, 10, 20 bars    | Exit rule for momentum variants.                                                                                            |
| Volume window         | 20, 40 bars       | Look-back window for average volume.                                                                                        |
| Volume ratio          | 0.8, 1.0, 1.2     | Entry filter for volume-confirmed variants.                                                                                 |

The selection rule prioritizes validation Sharpe ratio, then less severe drawdown and lower turnover. Zero-turnover candidates are not selected unless every candidate in the family does not trade.

The GARCH-Bollinger experiment is organized around two signal interpretations and three parameter sources. The two interpretations are momentum and contrarian. Momentum candidates are selected from GARCH-Bollinger momentum variants, with or without volume confirmation. Contrarian candidates are selected from standard Bollinger contrarian, GARCH-Bollinger contrarian, and GARCH-Bollinger contrarian with volume confirmation. The three parameter sources are original, day, and night. For each source, the best momentum setting and the best contrarian setting are selected on validation data only. In final testing, original-source parameters are evaluated on original, day, and night test sets, while day-source and night-source parameters are evaluated on their corresponding test sets. This creates five parameter-test settings and ten directional test rows, but all choices are fixed before the test results are observed.

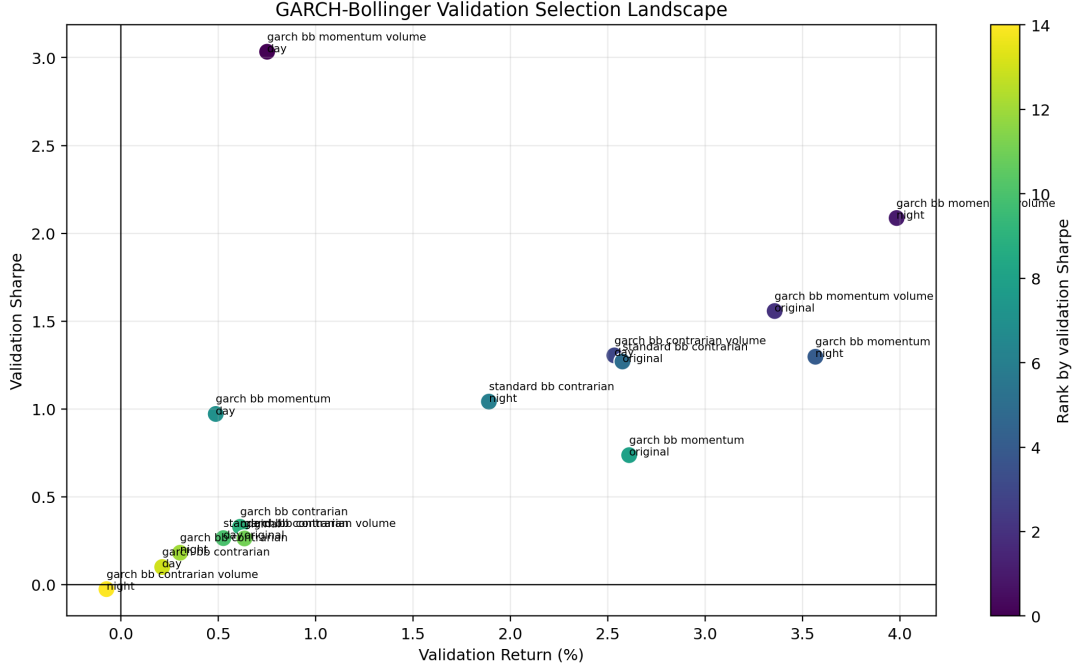


Figure 3: Validation selection landscape for GARCH-Bollinger candidates. The plot illustrates that validation performance is highly parameter- and regime-dependent.

The key methodological point is that test performance does not feed back into parameter selection. This protects the final test period from data snooping.

## 5 Backtesting, Slippage and Risk Management

### 5.1 Backtesting Assumptions

The backtests include transaction cost and slippage proxies. The common futures assumptions are:

Table 4: Main trading assumptions.

| Item                       | Value                  |
|----------------------------|------------------------|
| Initial capital            | HKD 5,000,000          |
| HSI futures multiplier     | HKD 50 per index point |
| Slippage proxy             | 1 index point          |
| GRPO transaction cost      | HKD 10 per contract    |
| LGBM transaction cost rate | 0.01% per side         |

The accounting conventions differ across implementations. GRPO reports HKD PnL using futures multiplier and initial capital. LGBM reports return-based metrics and also has a separate risk-management module. GARCH-Bollinger reports ratio returns for the technical strategy tests. Therefore, the results should be compared directionally rather than treated as perfectly identical measurement systems.

For GARCH-Bollinger, the backtest uses ratio returns rather than capital or wealth. The executed position is the previous bar’s target position, which prevents using a signal and its return in the same bar:

$$\text{ExecutedPosition}_t = \text{TargetPosition}_{t-1}, \quad (17)$$

$$\text{CloseReturn}_t = \frac{P_t}{P_{t-1}} - 1, \quad (18)$$

$$\text{CostReturn}_t = \frac{|\text{ExecutedPosition}_t - \text{ExecutedPosition}_{t-1}| \cdot \text{CostPointsPerSide}}{P_{t-1}}, \quad (19)$$

$$\text{StrategyReturn}_t = \text{ExecutedPosition}_t \cdot \text{CloseReturn}_t - \text{CostReturn}_t. \quad (20)$$

The current GARCH-Bollinger setting uses  $\text{CostPointsPerSide} = 1.0$ . Moving from flat to long has turnover 1, while flipping from long to short has turnover 2 because the old position must be closed and the opposite position opened.

## 5.2 LGBM Backtesting Results

The LGBM test results are strong across most scenarios, especially for the original model tested on original, day and night datasets.

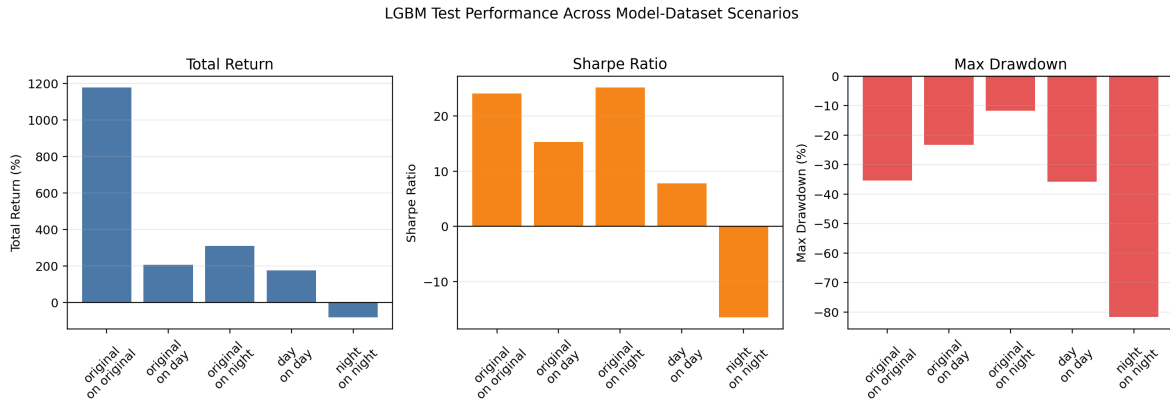


Figure 4: LGBM test performance by scenario. The original model generalizes better than the night-specific model in this experiment.

Table 5: LGBM test metrics.

| Scenario             | Total Return | Sharpe | Max DD  | Trades |
|----------------------|--------------|--------|---------|--------|
| Original on original | 1177.73%     | 24.07  | -35.44% | 6584   |
| Original on day      | 207.85%      | 15.29  | -23.35% | 4016   |
| Original on night    | 311.09%      | 25.13  | -11.80% | 2599   |
| Day on day           | 176.40%      | 7.78   | -35.81% | 4964   |
| Night on night       | -81.63%      | -16.51 | -81.67% | 7032   |



The naked (unconstrained) LGBM strategy exhibited exceptionally high cumulative returns exceeding on the full test period, with annualized returns reaching several thousand percent. However, these figures appear unrealistically inflated, primarily due to the high-frequency nature of the minute-level data and the resulting large annualization factor. When a comprehensive risk management framework — incorporating position sizing, volatility-adjusted exposure, daily drawdown limits, and consecutive-loss halts — was applied, performance became substantially more realistic, delivering a total return of approximately 40.1622% with a maximum drawdown of only -1.45% and win rate up to 66%. This highlights the critical importance of risk controls in translating theoretical model performance into practical, implementable results.

### 5.3 GRPO Backtesting Results

The fixed one-contract GRPO evaluation is weak before adding dynamic risk management. The least negative fixed case is `day_model` on `day_test`, but it still loses HKD 154,570 with Sharpe ratio -0.83.

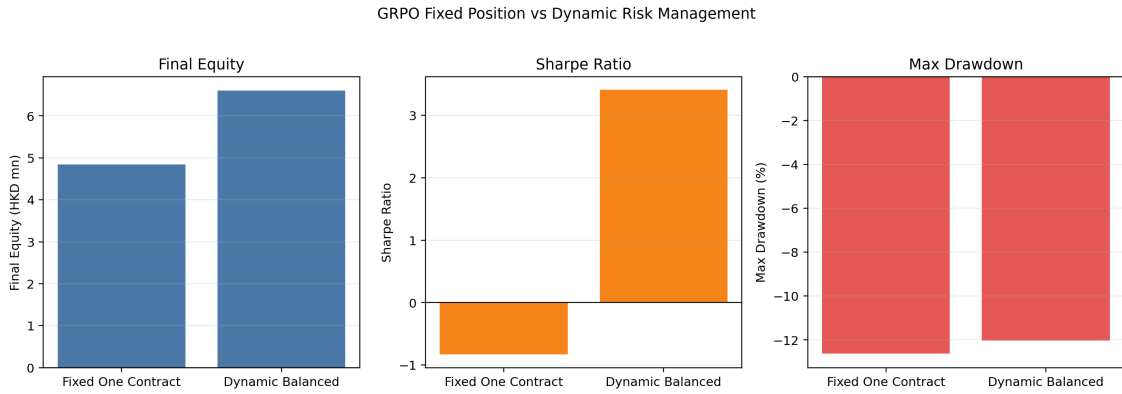


Figure 5: GRPO fixed one-contract baseline versus dynamic balanced risk management. Dynamic position sizing changes both return and risk characteristics.

Table 6: GRPO risk comparison.

| Method             | Final Equity | Total PnL | Sharpe | Max DD  | Trades |
|--------------------|--------------|-----------|--------|---------|--------|
| Fixed one contract | 4,845,430    | -154,570  | -0.83  | -12.63% | 7704   |
| Dynamic balanced   | 6,607,190    | 1,607,190 | 3.41   | -12.04% | 4188   |

The dynamic layer sizes positions using equity risk budget and recent volatility, reduces maximum contracts in protective drawdown mode, stops new trades under a kill switch, applies stop-loss and take-profit rules, and uses cooldown bars after stop-loss events.

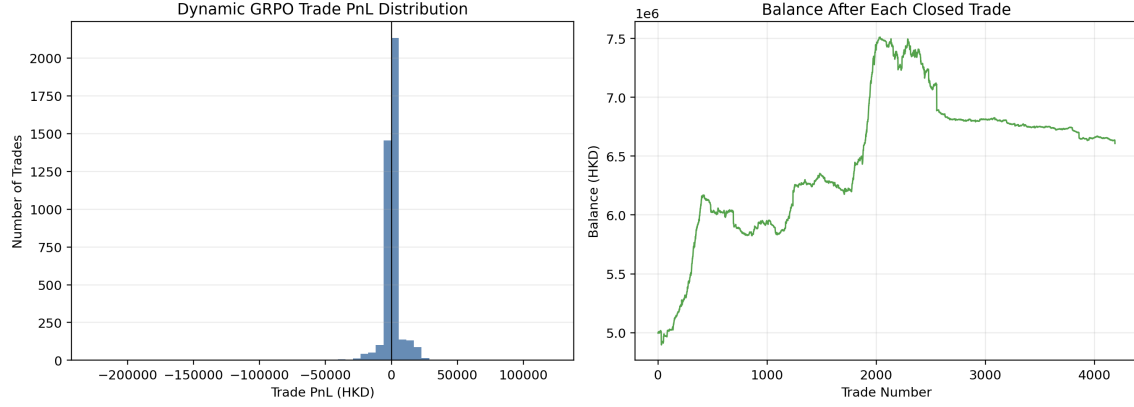


Figure 6: Dynamic GRPO trade PnL distribution and balance after each closed trade. The distribution shows frequent small trades with occasional large stop-loss events.

#### 5.4 GARCH-Bollinger Backtesting Results

For GARCH-Bollinger, we first use the validation set to select the best momentum setting and the best contrarian setting for each dataset. After the parameters are selected, they are fixed and tested on the out-of-sample test set. The table compares these fixed validation-selected settings. For each dataset combination, we report both the momentum result and the contrarian result, so the table has ten rows in total. Validation Sharpe is included to show the selection evidence, while test return and test Sharpe report final out-of-sample performance. The test set is used only for evaluation, not for selecting parameters.

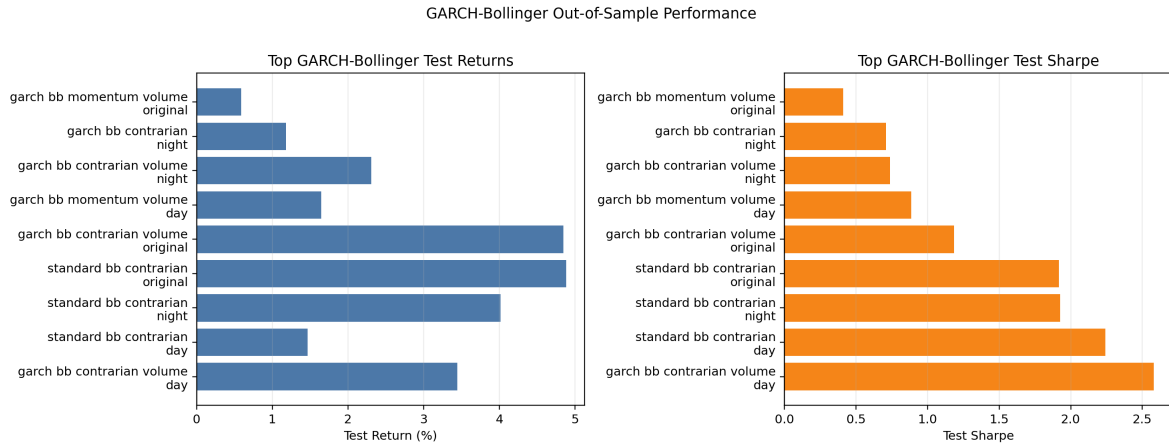


Figure 7: Top GARCH-Bollinger out-of-sample results. Contrarian variants dominate many of the strongest test cases.

Table 7: GARCH-Bollinger directional test results: five settings by two interpretations.

| Parameter source | Test set | Direction  | Validation-selected variant | Val. Sharpe | Test return | Test Sharpe | Max DD |
|------------------|----------|------------|-----------------------------|-------------|-------------|-------------|--------|
| Original         | Original | Momentum   | GARCH BB momentum volume    | 1.56        | 0.59%       | 0.41        | -1.45% |
| Original         | Original | Contrarian | Standard BB contrarian      | 1.27        | 4.88%       | 1.92        | -2.13% |
| Original         | Day      | Momentum   | GARCH BB momentum volume    | 1.56        | 0.00%       | 0.00        | 0.00%  |
| Original         | Day      | Contrarian | Standard BB contrarian      | 1.27        | 1.47%       | 2.24        | -0.58% |
| Original         | Night    | Momentum   | GARCH BB momentum volume    | 1.56        | 0.40%       | 0.34        | -1.45% |
| Original         | Night    | Contrarian | Standard BB contrarian      | 1.27        | 4.02%       | 1.93        | -2.13% |
| Day              | Day      | Momentum   | GARCH BB momentum volume    | 3.03        | 1.64%       | 0.89        | -1.97% |
| Day              | Day      | Contrarian | GARCH BB contrarian volume  | 1.30        | 3.44%       | 1.96        | -1.43% |
| Night            | Night    | Momentum   | GARCH BB momentum volume    | 2.09        | 0.58%       | 0.26        | -3.22% |
| Night            | Night    | Contrarian | Standard BB contrarian      | 1.04        | 4.02%       | 1.93        | -2.13% |

This table is the main GARCH-Bollinger finding. Momentum with volume confirmation had strong validation Sharpe, especially for day and night parameters, which suggests that the validation period rewarded breakout-style signals. However, in the final test period, contrarian rules delivered higher Sharpe ratios in all five directional settings. The result is therefore not simply that GARCH is always better than standard Bollinger Bands, or that momentum is always worse than contrarian trading. A more precise interpretation is that the validation period behaved more like a breakout environment, while the test period, especially late 2019 to early 2020, was more favorable to mean reversion after band breaches.

To provide a direct passive benchmark, Figure 8 compares the cumulative test-period performance of the three strategy lines with buy-and-hold (BAH) benchmarks. All curves are normalized to start from 1.0, so the figure focuses on relative cumulative performance rather than initial capital. The GARCH-Bollinger curve uses validation-selected contrarian parameters on the original test set. The LGBM curve uses the validation-selected risk-managed equity curve and is compared with BAH over the night-test period. The GRPO curve uses the dynamic-risk equity curve and is compared with BAH over its corresponding day-test period.

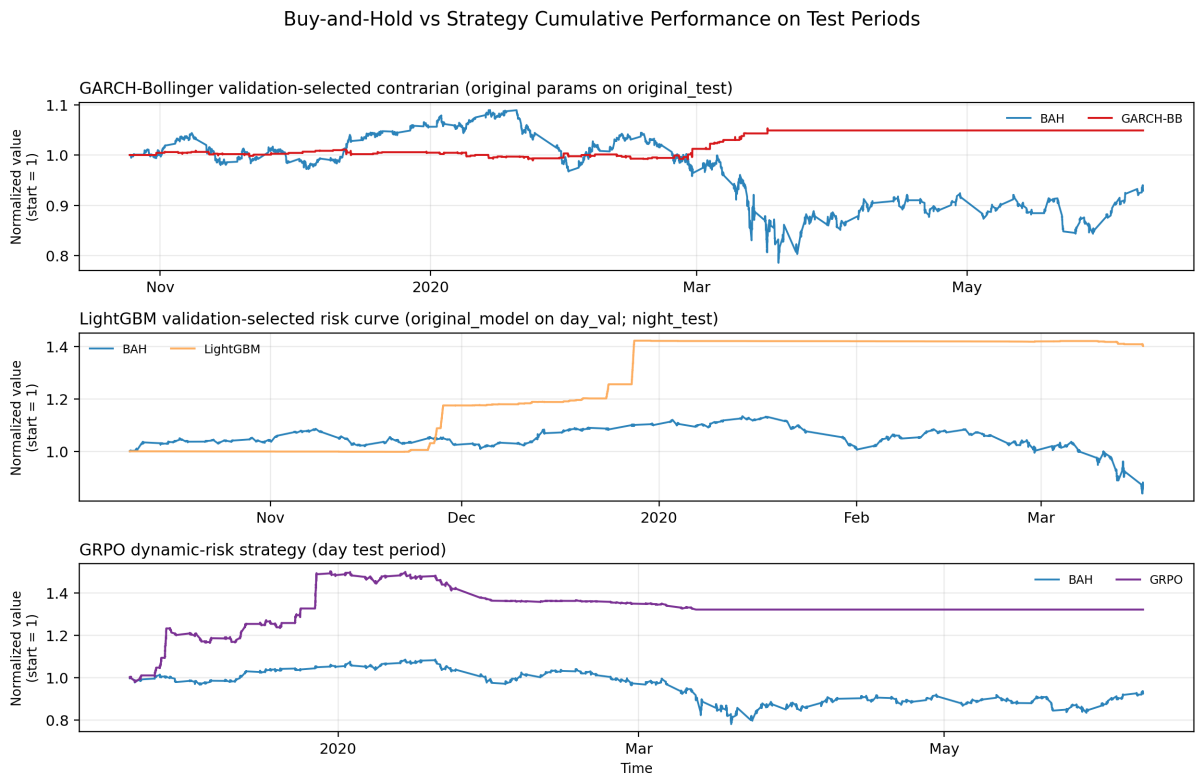


Figure 8: Buy-and-hold benchmark comparison for the three strategy lines on their corresponding test periods. All curves are normalized to start from 1.0.

The benchmark comparison shows that each active strategy outperforms its corresponding BAH curve in the selected out-of-sample evaluation. However, this figure should be interpreted as a benchmark check rather than a direct ranking across strategies, because the three curves come from different strategy designs, risk controls, and test-period definitions.

## 5.5 Slippage Discussion

The project models slippage with a simplified one-point proxy. In live trading, slippage depends on order type, queue position, bid-ask spread, market depth, volatility, news timing, and whether the strategy needs to reverse directly from long to short or short to long. Slippage can be larger during night sessions, market open and close, lunch break restart, macro news, overseas shocks, and large price gaps. Therefore, the backtest should be interpreted as a bar-level execution approximation, not a full order-book simulation.

## 5.6 Risk Management Features

The project implements risk management at several levels. For LGBM, the risk manager controls position sizing, maximum exposure, single-trade loss, daily drawdown, and consecutive-loss halt. For GRPO, the dynamic module uses risk budget, maximum contracts, protective drawdown, kill switch, stop-loss, take-profit, cooldown bars, volatility lookback, and minimum volatility assumptions. For GARCH-Bollinger, risk control is simpler but still present through band width, volume confirmation, maximum holding bars, and transaction-cost-adjusted returns.

Table 8: Main risk parameters for LGBM.

| Parameter                 | Value           |
|---------------------------|-----------------|
| Risk per trade            | 0.5% of equity  |
| Maximum contracts         | 5               |
| Protective drawdown       | 8%              |
| Kill-switch drawdown      | 12%             |
| Stop-loss multiple        | 1.0 risk budget |
| Take-profit multiple      | 1.5 risk budget |
| Cooldown bars             | 20              |
| Volatility lookback       | 60 bars         |
| Minimum volatility points | 20              |

Table 9: Main dynamic risk parameters for GRPO.

| Parameter                   | Value         |
|-----------------------------|---------------|
| max position ratio          | 30% of equity |
| max loss per trade          | 2%            |
| max drawdown daily          | 5%            |
| max num of consecutive loss | 5             |
| volatility per trade        | 1%            |

## 6 Real-Time Implementation Difficulties

The current project is a research and backtesting framework, not a production trading system. A live implementation would face several practical problems.

First, real-time data quality is harder than historical CSV backtesting. A live system must detect missing bars, delayed ticks, duplicated bars, lunch breaks, night-session boundaries, holidays, and abnormal timestamps. If live session handling differs from the backtest, rolling indicators, GARCH volatility, LGBM features, and GRPO observations may become inconsistent.

Second, signals should be generated only from completed bars. Using unfinished one-minute bars creates hidden look-ahead bias because high, low, close, and volume are not final yet.

Third, futures trading has contract lifecycle issues. HSI futures have expiry months, rollover dates, and liquidity migration from the expiring contract to the next active contract. A production system must decide when to roll and how to handle price gaps between contracts.

Fourth, margin and forced liquidation are central to futures trading. A broker imposes initial margin, maintenance margin, intraday checks, and liquidation rules. A sharp adverse move can trigger margin calls before the model’s intended stop-loss logic is reached.

Finally, a live system needs operational controls: kill switch, daily loss limit, order throttling, broker-position reconciliation, logging, alerting, and manual override.

## 7 Additional Discussion

The three strategies have different strengths and weaknesses. LGBM produces the strongest raw test performance, but the extremely high returns and Sharpe ratios should be interpreted carefully because intraday machine learning backtests are vulnerable to overfitting, subtle leakage, and optimistic execution assumptions. GRPO is conceptually attractive because it optimizes trading actions directly, but the raw fixed-contract policy is not profitable in this experiment. Its value appears only after dynamic position sizing and risk controls are added. GARCH-Bollinger is the most interpretable strategy. Its returns are less spectacular, but it clearly illustrates regime risk: validation favored momentum, while the final test period favored contrarian rules.

Overall, the project supports a cautious view of quantitative strategy development. A strong backtest is not enough. A credible trading strategy must also have a clear economic rationale, clean data splitting, realistic cost assumptions, robust out-of-sample testing, and risk management that can survive adverse market regimes.

## 8 Conclusion

This project compares supervised learning, reinforcement learning, and traditional volatility-based technical trading on intraday HSI futures data. The LGBM strategy achieves the strongest raw test performance, especially for the original model, but its results require careful interpretation because high-frequency prediction is sensitive to overfitting and execution assumptions. The unconstrained model demonstrated striking profitability in backtests, with total returns surpassing an extremely high result, such results should be interpreted with considerable caution. The exceptionally high annualized returns are likely amplified by the annualization methodology applied to minute-bar data, where even modest per-bar profitability is scaled by a large number of trading periods per year. In practice, these figures tend to overstate economic viability due to unmodeled frictions, capacity constraints, and execution realities. The GRPO strategy performs poorly under fixed one-contract trading, but the dynamic balanced risk-management layer significantly improves its final equity and Sharpe ratio. The GARCH-Bollinger strategy is more interpretable and shows that contrarian band-breach rules can be more robust than validation-selected momentum rules in the final test period.

The main lesson is that model complexity alone does not guarantee trading robustness. The final strategy quality depends on the full pipeline: data cleaning, chronological splitting, feature construction, validation discipline, transaction cost modeling, position sizing, drawdown

control, and implementation feasibility. For real deployment, the next steps should include more conservative execution simulation, walk-forward validation, live paper trading, and broker-level risk-control integration.

## 9 Code Repository

The project code and supporting files are available at: [PeaHaWf/8020GroupProject](#).

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